

TIMMINS, ON, Canada

WMO: 717390

Lat: 48.568N

Lon: 81.378W

Elev: 295

StdP: 97.83

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.2	-29.9	-37.4	0.1	-33.0	-34.2	0.2	-29.6	9.2	-12.8	8.4	-13.9	1.6	280	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	29.5	19.8	27.6	18.5	25.9	17.8	21.4	27.0	20.2	25.4	19.1	23.9	4.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.8	23.9	18.4	13.8	22.9	17.3	12.8	21.7	63.8	27.2	59.6	25.4	55.8	24.0	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.7	6.6	DB	-37.7	32.7	3.3	1.9	-40.0	34.1	-42.0	35.2	-43.8	36.3	-46.2	37.7	(4)
(5)			WB	-37.8	23.5	3.3	1.0	-40.1	24.2	-42.0	24.8	-43.9	25.4	-46.3	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-16.1	-13.9	-7.1	1.1	9.7	15.4	17.7	16.3	12.1	5.0	-2.9	-10.9	(6)	
	DBStd	13.15	8.08	7.25	7.55	5.75	5.16	4.46	3.53	3.67	4.78	4.98	6.14	7.42	(7)	
	HDD10.0	3600	809	671	532	271	71	8	0	1	31	173	387	647	(8)	
	HDD18.3	5948	1068	904	789	517	272	108	55	84	194	414	637	905	(9)	
	CDD10.0	784	0	0	1	4	61	170	239	198	94	17	1	0	(10)	
	CDD18.3	90	0	0	0	0	4	21	36	23	7	0	0	0	(11)	
	CDH23.3	1064	0	0	1	3	79	288	392	227	73	2	0	0	(12)	
	CDH26.7	266	0	0	0	1	19	81	102	49	14	0	0	0	(13)	
	WSAvg	3.3	3.3	3.5	3.6	3.7	3.5	3.1	2.8	2.8	3.0	3.3	3.4	3.3	(14)	
(15) Precipitation	PrecAvg	921	60	49	66	53	75	97	100	96	95	75	82	69	(15)	
	PrecMax	1130	145	104	152	108	142	220	194	206	176	162	178	121	(16)	
	PrecMin	756	18	13	20	8	22	27	45	26	56	38	20	28	(17)	
	PrecStd	98	26	23	31	21	30	41	36	40	28	27	35	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	6.3	15.2	21.5	29.5	31.9	32.1	30.6	28.5	22.5	14.1	4.7	(19)	
		MCWB	2.2	4.0	9.6	11.4	17.8	20.4	21.6	21.0	19.7	15.4	9.8	3.4	(20)	
	2%	DB	0.8	2.5	9.7	16.7	25.3	28.9	29.3	27.9	25.3	18.9	10.4	2.5	(21)	
		MCWB	0.2	0.9	5.3	8.6	15.0	19.1	20.0	19.6	18.4	13.8	8.1	1.6	(22)	
	5%	DB	-0.8	-0.3	6.4	13.4	22.6	26.6	27.3	25.9	22.9	15.9	7.6	1.1	(23)	
		MCWB	-1.5	-1.6	2.8	6.7	13.7	17.1	18.7	18.7	17.1	12.4	5.7	0.3	(24)	
	10%	DB	-3.3	-2.8	3.6	10.2	19.9	24.4	25.4	23.8	20.5	13.1	5.2	-0.3	(25)	
		MCWB	-4.0	-3.9	0.8	5.0	12.4	16.3	18.1	17.5	16.0	10.3	3.8	-0.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	4.2	11.1	13.2	20.4	22.2	22.9	22.5	21.2	17.1	11.0	4.0	(27)	
		MCDB	3.0	5.6	13.7	18.9	25.1	29.6	29.6	27.8	25.9	20.5	12.7	4.5	(28)	
	2%	WB	0.3	0.7	5.8	9.4	17.1	20.5	21.4	20.8	19.5	14.7	8.5	1.7	(29)	
		MCDB	0.7	2.4	8.8	14.8	23.0	26.5	26.9	25.9	23.7	17.9	10.1	2.4	(30)	
	5%	WB	-1.6	-1.6	3.1	7.4	14.8	19.0	20.2	19.6	18.1	12.7	5.9	0.4	(31)	
		MCDB	-0.8	-0.5	6.2	12.5	20.1	23.9	25.3	24.4	21.8	15.9	7.3	1.0	(32)	
	10%	WB	-4.0	-3.8	1.0	5.4	13.0	17.5	19.0	18.4	16.6	10.4	3.8	-0.9	(33)	
		MCDB	-3.2	-2.8	3.4	9.9	19.0	22.2	23.9	22.8	20.0	13.0	5.3	-0.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	12.1	12.6	11.6	13.3	13.1	12.5	11.9	11.3	8.5	7.2	8.5	(35)	
		MCDBR	9.5	10.8	13.6	17.1	18.0	16.7	15.1	14.8	14.3	12.9	9.8	7.5	(36)	
	5% WB	MCWBR	9.1	9.6	9.7	10.4	9.7	8.2	7.3	7.6	8.2	8.5	7.8	7.2	(37)	
		MCWBR	9.2	10.5	11.9	15.2	14.7	13.2	12.8	12.5	12.4	11.6	9.1	7.2	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.274	0.271	0.299	0.341	0.358	0.377	0.391	0.375	0.360	0.323	0.287	0.258	(40)	
		taud	2.207	2.246	2.257	2.323	2.417	2.370	2.363	2.405	2.433	2.521	2.483	2.331	(41)	
	Edn at Noon	Edn at Noon	780	887	916	908	903	884	864	865	843	825	785	768	(42)	
		Edh at Noon	78	94	111	117	111	117	117	107	94	73	60	60	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.28	2.35	3.80	4.72	5.22	5.64	5.42	4.55	3.40	1.99	1.06	0.84	(44)	
	RadStd	RadStd	0.21	0.31	0.37	0.41	0.43	0.36	0.48	0.27	0.27	0.27	0.11	0.12	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page